Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/26/2005
PI Name:	Landry, Donald M.D., Ph.D.		
Project Title:	Translation of Disease Markers into Bioluminescent Signals		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	New York	**State:**	NY
Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/01/2002	**End Date:**	12/31/2004
No. of Post Docs:	4	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:	Fung, Paul	**Contact Phone:**	650-604-1276
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Landry Donald		
COI Name (Institution):	Stojanovic, Milan (Columbia University)		
Grant/Contract No.:	NAS2-02039		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are developing autonomous devices, based on networks of nucleic acid sensors and enzymes. These devices will be able to home in to specific tissues, recognize multiple disease markers, and that recognition will trigger the release of small molecules that can be analyzed in urine. Our long-term objective is to achieve the non-invasive continuous following of astronauts health during long flights. Our short term objective is to develop enabling technologies: (1) the ability to release a small molecule upon a molecular recognition event; (2) decision-making networks capable of analyzing multiple signals and performing Boolean calculation resulting in a release of small molecule; (3) arrays of sensors capable of continuous following of small molecules in urine. Our results will give impetus to further research in the emerging field of autonomous diagnostic and therapeutic devices. Short-term, we will develop useful cross-reactive arrays for the analysis of steroids in urine and new types of targetable therapeutics.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have essentially accomplished the most important task of our project, along the lines of the initial RFA: We obtained a proof-of-principle for a technology capable of autonomously analysing multiple disease markers and releasing small molecules, either for reporting or treatment purposes. For the first time ever, we have a set of molecules capable of complex decision-making in solution, resulting in a decision to release or not a drug. This technology opens up new possibilities in “physician-less” diagnostic procedures at molecular level, and encourages us to pursue technical improvements that would lead to practical applications (e.g. rate increase, stability issues). Several technologies with immediate practical applications were also developed. For example, we are running clinical trials with cross-reactive arrays for steroid analysis in urine, and we are testing our ability to image intracellular processes with modular aptameric sensors.
Bibliography Type:	Description: (Last Updated: 07/26/2005)
Articles in Peer-reviewed Journals	Stojanovic, M., N.; Green, E. G.; Semova, S.; Nikic, D. B. & Landry D. W. "Cross-reactive Arrays Based on Three-Way Junctions" J. Am. Chem. Soc. 125, 6085-6089 (2003) , May-2003
Articles in Peer-reviewed Journals	Stojanovic M.N.*; Landry D.W. "Aptamer-based colorimetric sensor for cocaine" J. Am. Chem. Soc. 124: 9678, (2002) , Jan-2003
Articles in Peer-reviewed Journals	Stojanovic M. N.*; Stefanovic D. "Deoxyribozyme-based half-adder" J. Am. Chem. Soc. 125: 6637, (2003) , May-2003
Articles in Peer-reviewed Journals	Stojanovic, M., N. & Kolpashchikov, D. M. "Modular Aptameric Sensors" . Am. Chem. Soc. 126, 9266-9270 (2004) , Aug-2004
Articles in Peer-reviewed Journals	Stojanovic, M., N. & Stefanovic D. "Deoxyribozyme-based automaton" Nature Biotechnol. 21, 1069-1074 (2003) , Sep-2003